

TORSIONVERSE: COSMOLOGICAL REDSHIFT — CANDIDATE MECHANISMS AND VOID-EDGE PREDICTION

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TORSIONVERSE: COSMOLOGICAL REDSHIFT — CANDIDATE MECHANISMS AND VOID-EDGE PREDICTION

Companion script: analysis/demos/redshift_doc.py -- 23/23 PASS [RS1-RS15, RS3a-c]

Secondary: analysis/demos/cell_expansion_age.py -- 3/3 PASS [TV-native age RS3b-d]

Secondary check: analysis/demos/redshift_transfer_check.py -- 5/5 PASS [TC1-TC5]

(Exploratory framing checks -- these confirm the numbers are computed correctly with zero free parameters, NOT that any candidate mechanism is proven. This entire doc is open research, not a closed derivation.)

ABSTRACT

In the torsionverse framework, a photon is a localized T_{1g} corpuscle that travels at speed c through the Jobson cell medium. The local medium has a characteristic frequency $f_{\text{cell}} = c / L_J_{\text{local}}$, where L_J_{local} is the local Jobson cell spacing. What the observer measures is this local medium frequency -- a medium property, not an intrinsic property of the corpuscle. Because $L_J \propto n_{\text{cells}}^{-1/3}$ (cell spacing grows as cells become sparser), the local medium frequency at a sparse void edge is lower than in a dense filament:

$$\begin{aligned} f_{\text{obs}} / f_{\text{emit}} &= (n_{\text{void_edge}} / n_{\text{filament}})^{1/3} \\ z + 1 &= (n_{\text{filament}} / n_{\text{void_edge}})^{1/3} \quad [\text{Candidate D, RS12}] \end{aligned}$$

The corpuscle speed (c) is unchanged. The scale argument rules out the per-cell interaction picture: $\lambda_{\text{visible}} / L_J_{\text{dense}} \approx 5 \times 10^{10}$. The wave is fifty billion times larger than one cell; the medium is continuous from the wave's perspective in both dense filaments and the deepest observed voids ($\lambda / L_J_{\text{void}} \approx 2 \times 10^{10}$ at 10% cell density). No individual cell resonance is physically accessible to any observable photon. The mechanism is a collective medium property, not a per-cell interaction.

$f_{\text{cell}} = c / L_J_{\text{local}}$ is the local medium's characteristic frequency -- the local propagation parameter set by cell spacing. The precise mechanism connecting local cell density to spectroscopic redshift (boundary frequency shift, accumulated phase, or medium clock rate) is an open derivation: Series 3 target.

In sparse voids, ambient radiation pressure from all directions holds cells at their equilibrium positions. This is a mechanical effect on cell location,

independent of EM wave propagation speed, which is set by $\epsilon_{\text{eff}} / \mu_{\text{eff}}$.

Radiation pressure is ~25 orders of magnitude weaker than EM stiffness

($P_{\text{CMB}} / (1/\epsilon_0) \approx 1e-25$) and does not contribute to c .

This mechanism (a) is consistent with c constant everywhere (RS11),

(b) predicts a Hubble-tension correction of ~7-9% for the KBC void (25%

local underdensity), (c) predicts T_{1g} amplitude enhancement at cell density

boundaries -- consistent with unexplained heat observations at the heliosphere

boundary (RS13), and (d) identifies the Hubble radius c/H_0 as the cell

propagation horizon beyond which no medium exists and therefore no structured matter can survive (RS15).

The framework examines four candidate redshift mechanisms:

- (A) Standard metric expansion (currently imported from Λ CDM, not derived)
- (B) "Local boson emission" (referenced elsewhere, never developed -- stale)
- (C) Pure cell migration outward -- examined here for the first time; ruled out as a creation mechanism but established as pure migration (RS7a)
- (D) Sparse void edges $\rightarrow$ lower local medium frequency $\rightarrow$ redshift -- the primary new result; quantified with zero free parameters (RS12-RS15)

Beyond the cell propagation horizon, matter loses structural coherence (Hopf

windings require cells as substrate). Communication across the boundary remains

possible in principle if the sender corrects for the clock-rate shift, but

physical travel is prohibited by substrate dissolution, not by energy or speed.

All 20 checks pass [RS1-RS15] as exploratory framing checks; this paper is open research, not a closed derivation of the cosmological constant or expansion rate.

SECTION 1: FOUR REDSHIFT MECHANISM CANDIDATES

(A) STANDARD METRIC EXPANSION [currently used, not derived]

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H(z) = H0 * sqrt(Omega_m*(1+z)^3 + Omega_Lambda) [Planck 2018 LambdaCDM]
a0(z) = Rs * c * H(z) [analysis/gravity/a0_redshift.py]
a0(z=1) = 2.086e-10 m/s^2 (79% higher than today) -- JWST-testable.
STATUS: H0 and the expansion history are IMPORTED from standard
cosmology. The torsionverse contributes only the Rs*c normalization
that converts H(z) into a MOND-type acceleration scale. The mechanism
of expansion itself (why space/the medium expands at all) is NOT
addressed by this framework -- it is inherited unmodified from FRW/GR.
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(B) "LOCAL BOSON EMISSION" [referenced, never developed -- STALE]

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doc_torsion.txt (line ~872) states: "An alternative interpretation being
investigated (doc_orbit_pressure.txt) treats redshift as locally
generated by stellar/planetary boson emission, in which case H would
be approximately constant across z and a0 would be constant."
CONFIRMED: doc_orbit_pressure.txt contains NO
such development. This is a stale forward-reference to work that was
never done. It remains a nominally distinct candidate (redshift as a
cumulative local emission/absorption process along the photon's path,
similar in spirit to historical "tired light" proposals) but has zero
quantitative content in the current framework.
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(C) CELLS MIGRATING OUTWARD INTO EMPTY MEDIUM VOLUME [new hypothesis]

Physical picture: the Jobson cell medium is not a passive coordinate grid that "stretches" -- it is a physical elastic substance. If new volume is continuously becoming available (however that happens), existing cells could either (i) dilute in place (spread thinner, same total cell count, standard FRW matter dilution $\sim 1/a^3$), or (ii) new

cells could be continuously created to hold density constant, with the medium's rest density ρ_{Λ} staying fixed by ongoing creation rather than by (i)-style dilution of a fixed-count medium. ρ_{Λ} (cosmological dark-energy-like density) IS observed to stay approximately constant as the universe expands -- which under naive FRW energy accounting already implies something like case (ii) for the dark-energy component, independent of any torsionverse-specific claim. STATUS: not examined in the literature prior to this paper. See Section 2 for a first quantitative pass.

(D) VOID PROPAGATION -- REDSHIFT FROM SPARSE VOID EDGE CELLS [new hypothesis]

Physical picture: the KEY ZONE is the void EDGE: where cell density transitions sharply from galaxy-filament density to void density. Whether void interiors are cell-free (swept empty by net radiation pressure from surrounding filaments) or merely sparse is an open observational question -- cell bridges between stars may persist along filamentary paths. Regardless, the density gradient at the edge is the operative mechanism: a photon traversing this transition propagates through a medium with increasing L_J local, and the observer in dense medium interprets the longer-scale T_{1g} disturbance as longer wavelength = lower frequency = REDSHIFT.

MECHANISM:

In dense medium: local medium characteristic period $\tau_{\text{dense}} = L_J / c$
 In sparse void edge (density $n_{\text{void_edge}}$): $L_{J_void} = L_J \times (n_{\text{dense}}/n_{\text{void_edge}})^{1/3}$
 $\tau_{\text{void}} = L_{J_void} / c = \tau_{\text{dense}} \times (n_{\text{dense}}/n_{\text{void_edge}})^{1/3}$
 The sparse-edge medium has a longer characteristic period (slower clock).
 Photons emitted in (or passing through) the slow-clock region appear redshifted when measured by the observer's faster dense-region clock:

$$z + 1 = \tau_{\text{void}} / \tau_{\text{dense}} = (n_{\text{dense}} / n_{\text{void_edge}})^{1/3}$$

RS11 COMPATIBILITY: c is unchanged.

The redshift arises from the local clock rate (medium characteristic period), not from a change in propagation speed. RS11 applies to speed; this mechanism applies to the ratio of local to observer clock rates.

QUANTITATIVE ESTIMATE (RS12 -- see below):

For void edge with $n_{\text{void_edge}} / n_{\text{dense}} = 0.75$ (25% underdensity):
 $z_{\text{per_crossing}} = (1/0.75)^{1/3} - 1 = 1.333^{1/3} - 1 \approx 0.101$
 For a path through multiple void crossings ($f_{\text{void}} = 30\%$ of path):
 $z_{\text{total}} \approx f_{\text{void}} \times 0.101 \times d / L_{J_void} \approx \text{Hubble-like scaling}$
 KBC void (25% local underdensity): $H0_{\text{local}} / H0_{\text{CMB}} \approx (0.75)^{1/3} \approx 0.91$
 $\rightarrow \sim 9\%$ Hubble tension in correct direction.

OPEN ITEMS:

- Formal derivation of $z = (n_{\text{dense}}/n_{\text{void}})^{1/3} - 1$ from torsionverse medium geometry (local clock = medium characteristic period = L_J local / c)
- Quantitative fit: does void cell density from SDSS/DES void catalogs reproduce $z \propto d$ (Hubble law) when convolved with line-of-sight structure?
- RS12 check: compute z_{total} vs d from observed void distributions and compare to measured Hubble law. This is scriptable.

INTERNAL CONSISTENCY CHECK (RS11 -- still holds):

From RS9: $B_{\text{eff}}/\rho = c^2$ everywhere. c is constant. The redshift arises from clock-rate differences (medium characteristic period), not speed changes. D2 ($c_{\text{void}} \neq c_{\text{cells}}$) remains ruled out. [RS11 PASS]

CONSEQUENCE: HARD PHYSICAL BOUNDARY FOR MATTER AND TRAVEL

The cell propagation horizon ($c/H0 \approx 14.5$ Gly, RS15) is not merely an information horizon -- it is a hard physical substrate boundary.

Matter in torsionverse = Hopf windings in Jobson cells (Series 1, derived).
 Photons = localized T_{1g} wave packets propagating through the Jobson cell medium (derived).
 Both require cells to exist. Beyond the cell horizon: no cells $\rightarrow$ no medium
 $\rightarrow$ no Hopf windings $\rightarrow$ matter cannot maintain structural coherence.

An electron approaching the cell boundary would find progressively fewer cells to sustain its (1,2) winding. At the boundary, the winding unravels: the energy stored in the winding is no longer a stable topological

configuration – there is nothing to wind around. The electron ceases to exist as an electron. Its constituent energy dissipates into... nothing that any medium-based physics can describe.

This is fundamentally different from a Schwarzschild event horizon:

- GR event horizon: matter can cross (in principle); only communication is cut. The boundary is a coordinate singularity.
- Cell propagation horizon: matter CANNOT maintain structure beyond the boundary. The boundary is a physical substrate limit.

Safe travel implication: no craft can travel beyond the cell horizon and remain as structured matter. The cells that constitute the craft – and the Hopf windings (atoms, molecules) within it – would lose their substrate. This is not a velocity or energy limit; it is a structural impossibility. The universe has a hard boundary not because of curvature but because the medium that makes matter possible simply ends there.

FERMI PARADOX COROLLARY:

Every civilization in the observable universe faces the same hard physical boundary for matter travel. No craft – regardless of technology – can cross the cell propagation horizon and remain as structured matter. If other cell islands exist beyond our Hubble radius, their matter cannot physically reach ours.

However, COMMUNICATION across cell density gradients is not absolutely forbidden. A civilization that understands the Candidate D clock-rate mechanism ($z + 1 = (n_{\text{dense}}/n_{\text{void}})^{1/3}$) could pre-compensate their signal: by adjusting the transmission frequency upward by the void clock factor, the signal arrives at the observer's end at the correct local frequency. Cross-island communication would require:

- (1) Knowledge of the cell density structure between sender and receiver
- (2) A signal channel that survives propagation through sparse cell regions (T_{2g} gravitational waves propagate through sparser medium than T_{1g})
- (3) Patience: round-trip communication time = $2 \times \text{distance} / c$

The silence from beyond our cell island may reflect the bootstrap problem (two sides must agree on the correction factor before either can send a correctable signal), not a fundamental prohibition on communication. Within our shared cell region, standard Fermi constraints still apply locally (distance, time, RS13 void signal attenuation).

SECTION 2: CELL-GENERATION-RATE CROSS-CHECK (Section 2, RS1-RS8)

Companion: analysis/demos/redshift_doc.py

This section uses H₀ (observed, Planck 2018) to derive the required creation rate.

Note: the OLD RS3 used rho_Lambda (dark energy density -- external import) for n_cells; this was contaminated. The TV-native calculation (RS3a-c) uses $n_{\text{cells}} = 1/L_J^3$.

If cell density is held constant by continuous creation (candidate C, case ii):

Fractional volume growth rate: $dV/V/dt = 3 \cdot H_0 = 6.553e-18 \text{ /s}$ [RS2]
(This is a pure dimensional identity for ANY density-preserving creation hypothesis -- the same result appears in Bondi-Gold-Hoyle steady-state cosmology for continuous hydrogen creation. It carries no new physics by itself; the "3" comes from $V \sim r^3$.)

TV-NATIVE creation rate (RS3a-c, no rho_Lambda):

$n_{\text{cells}} = 1/L_J^3 = 1.02e+51 \text{ cells/m}^3$ (from geometry alone)
 $R_{\text{geo_TV}} = \text{gamma_flux} \times L_J^2 \times n_{\text{cells}} = 1.0e+16 \text{ cells/(m}^3 \text{ s)}$ [RS3a]
 $H_{\text{TV}} = R_{\text{geo_TV}} \times L_J^3 / 3 = 3.3e-36 \text{ /s}$ (TV-native expansion rate)
TV-native timescale >> 13.8 Gyr by many orders of magnitude [RS3b]
Time to reach 120 AU from 1 AU at H_TV consistent with heliosphere [RS3c]
NOTE: exact age depends on local vs cosmological flux and on whether cloning or migration dominates. The safe conclusion: if cloning drives expansion, our cell island is significantly older than the mainstream estimate. The specific number is not claimed without more supporting data.

LEGACY RS3 (rho_Lambda-contaminated, for reference):

$n_{\text{cells}} = \rho_{\text{Lambda}} c^2 / E_{\text{cell}} = 2.626 \times 10^{-02} \text{ cells/m}^3$ [CONTAMINATED]
Creation rate = $n_{\text{cells}} * 3H_0 = 1.721 \times 10^{-19} \text{ cells/(m}^3 \text{ s)}$ [legacy RS3]

Comparison to τ_{relax} (medium mechanical relaxation timescale, wave_dispersion.py lower bound $1e8 \text{ s}$):

$1/(3H_0) = 1.526 \times 10^{17} \text{ s} = 4.8 \text{ Gyr}$

$\tau_{\text{relax}} = 1e8 \text{ s} = 3.2 \text{ yr}$

Ratio: 1.53×10^9 -- these differ by 9.2 orders of magnitude. [RS4]

If cell creation (were it real) were mechanically tied to the medium's own relaxation process, these timescales would be expected to be comparable. They are not, by 9 orders of magnitude. This is a MILD argument against "continuous creation at the medium's natural rate" -- but τ_{relax} describes quasi-static flow response to a SUSTAINED external field (Section 6, doc_cell_depletion.txt), which has no established connection to a hypothetical cosmological creation process. This comparison is exploratory, not a proof either way.

Thermal pair-production at CURRENT CMB temperature (2.725 K) was tested as a candidate "cell generation" mechanism and RULED OUT: the Boltzmann suppression $\exp(-2*m_e*c^2/kT_{\text{CMB}})$ has exponent 4.35×10^9 -- the rate is zero for all practical purposes today. [RS5] Whatever "cell generation" might mean cosmologically, it is NOT ordinary thermal pair production at the present CMB temperature.

WHAT WOULD ACTUALLY CLOSE THIS [RS7, the concrete next steps]:

- A physical hypothesis for HOW cell creation (if real) couples to the expansion rate. Section 2's rate is a CONSEQUENCE of assuming constant ρ_{Lambda} ; it is not a derivation of any creation mechanism. Nothing in the framework currently explains WHY new cells would appear, only what rate would be required IF they do.
- A distinguishing observational test between candidate A (metric expansion) and candidate C (cell migration): does candidate C predict any deviation from the standard luminosity-distance/redshift relation, BAO scale, or CMB acoustic peaks? Not yet worked out.
- Formalizing "cells moving outward" as an actual solution of the medium's own wave equation, and checking consistency with $v_p=c$ (GW170817) and $v_s=Rs*c$ (flyby anomaly) -- both already fixed by other, unrelated measurements. Not yet attempted.

PRIORITY: HIGH. This is the most significant open gap identified in this

review -- the framework makes a specific falsifiable JWST prediction

($a_0(z=1)=2.086 \times 10^{-10} \text{ m/s}^2$) built entirely on an UNEXAMINED assumption about

what "the universe expands" physically means for an elastic medium made of

discrete cells. Closing (a)-(c) above should be the next major push.

SECTION 5: OPEN ITEMS

GENUINELY OPEN:

- Observational test distinguishing candidate A (metric expansion) from candidate C (cell migration) -- luminosity distance, BAO, CMB acoustic peaks (RS7b). Requires comparison with DESI/JWST datasets.
- Candidate D formal derivation: $z = (n_{\text{dense}}/n_{\text{void}})^{(1/3)} - 1$ from torsionverse lattice geometry (local clock = medium characteristic period = L_J_{local} / c). Quantitative fit to observed void catalogs (SDSS/DES) needed to confirm whether the mechanism reproduces $z \propto d$ (Hubble law). [RS12 scripted for 25% underdensity estimate; full survey fit is the next step]
- Formalizing the "all times exist as standing frequencies" hypothesis as an actual wave-equation solution and testing it against v_p , v_s (Section 4).
- The stale doc_torsion.txt forward-reference to a "local-boson emission" redshift alternative requires a future erratum to doc_torsion.txt.

ESSENTIALLY CLOSED:

- [x] L_J is a single-valued medium property, not a per-wave tunable parameter -- geometric argument against naive frequency-retuning to a different medium at the same location [Section 3, RS8]

- [x] Thermal pair production at current CMB temperature is not a viable

"cell generation" mechanism -- ruled out by Boltzmann suppression [RS5]

[x] Stellar photons CANNOT replenish a depleted zone. To create one new Jobson cell requires $E \geq E_{\text{cell}} = 124.8 \text{ GeV}$. Solar/stellar photons are a few eV -- 13 orders of magnitude too weak. [RS9a]

[x] Sustained intense EM fields CAUSE cell depletion, not replenishment. AGN at galactic center actively depletes cells. [RS9b]

[x] RS7a (WHY cells are created/migrate): CLOSED AS PURE MIGRATION. CELL CLONING CONDITION derived: a photon at $m_{\text{crit}} = 9.933 \text{ GeV}$ hitting a cell face at normal incidence satisfies both the Bragg condition and the winding nucleation condition simultaneously (exact algebraic identity: $\hbar c / (2 L_J) = m_p / (8 \alpha \phi) = m_{\text{crit}}$). This is mathematically elegant but DOES NOT explain the cosmological cell density.

RATE CHECK: cosmic gamma-ray flux at 10 GeV (Fermi-LAT IGRB $\sim 0.1 \text{ m}^{-2} \text{ s}^{-1}$) gives a geometric encounter rate $R_{\text{geo}} = 1.8 \times 10^{-37} \text{ cells}/(\text{m}^3 \text{ s})$. Required rate = $1.72 \times 10^{-19} \text{ cells}/(\text{m}^3 \text{ s})$. Shortfall = 22 orders of magnitude. Clone probability $P > 10^{17}$ required -- physically impossible.

CONCLUSION: GRBs/supernovae CANNOT create cells at the required rate. Candidate C = PURE MIGRATION (no new creation). All cells were created in the hot early universe at $T > E_{\text{cell}}/k_B \sim 10^{15} \text{ K}$ ($t < 10^{-8} \text{ s}$). Since then: no new cells, density slowly decreasing as volume expands. Dark energy density IS dropping over cosmic time. Consistent with DESI 2024 w0-wa parametrisation showing possible $w \neq -1$ (dynamical dark energy). The cell creation rate RS3 ($1.72 \times 10^{-19} \text{ cells}/(\text{m}^3 \text{ s})$) is NOT what's happening -- that was the required rate IF density is constant. It is not.

[x] RS7c: Wave-equation consistency of candidate C (pure migration): $v_s = \sqrt{G_{\text{shear}}/\rho} = \sqrt{\rho/(R_s c)^2/\rho} = R_s c$ exactly. The ρ cancels; v_s is independent of cell density. As cells migrate and density slowly decreases, the wave speeds $v_p=c$ and $v_s=R_s c$ remain exactly constant. Candidate C is wave-equation consistent. [RS9 PASS]

[x] Confirmed the doc_torsion.txt "local boson emission" cross-reference is stale (never developed in doc_orbit_pressure.txt)

[x] Candidate D mechanism revised (session 15-16): Original D1 (empty voids $\rightarrow$ transparent) and D2 (c changes $\rightarrow$ ruled out) were both incomplete. Voids are SPARSE, not empty. The revised mechanism (sparse void edge $\rightarrow$ local clock redshift, $z=(n_{\text{dense}}/n_{\text{void}})^{(1/3)-1}$) and heliosphere / cell-island boundary predictions (RS12-RS15) are in the script. Formal derivation from lattice geometry still open (RS12).

COMPANION SCRIPT

analysis/demos/redshift_doc.py -- 23/23 PASS [RS1-RS15, RS3a-c]
 analysis/demos/cell_expansion_age.py -- 3/3 PASS [RS3b-d TV-native age]
 (Exploratory framing checks: confirms numbers are computed correctly with zero free parameters. Does not itself prove any redshift mechanism.)

EXTERNAL INPUTS:

$H_0 = 67.4 \text{ km/s/Mpc}$, $\Omega_m=0.315$, $\Omega_{\Lambda}=0.685$ [Planck 2018]
 $T_{\text{CMB}} = 2.7255 \text{ K}$ [standard measurement]
 γ_{flux} at $\sim 10 \text{ GeV} = 0.1 / \text{m}^2/\text{s}$ [Fermi-LAT IGRB]
 NOTE: ρ_{Λ} (dark energy density) was used in legacy RS3 -- contaminated.
 TV-native RS3a-c uses $n_{\text{cells}} = 1/L_J^3$ instead.

TORSIONVERSE-DERIVED:

$E_{\text{cell}} = 124.8 \text{ GeV}$, $L_J = \alpha \phi r_p$, $\tau_{\text{relax}} > 1 \text{e8 s}$ [wave_dispersion.py]
 $R_s = \sqrt{5}/(4 \pi)$ [medium saturation constant]

NOTE ON PLACEMENT: See doc_series2_synthesis.txt companion papers list.